

# ANKLE REHABILITATION DEVICE TAXONOMY

Backup summary emphasising where the present thesis is positioned in the broader design space.

|                  | DEPLOYMENT                             | MECHANICAL EMBODIMENT                                   | ACTUATION LEVEL                                    |
|------------------|----------------------------------------|---------------------------------------------------------|----------------------------------------------------|
| Platform-based   | Large external rigs<br>clinic-oriented | Single-joint modules<br>or test platforms               | Electric motors<br>pneumatic systems               |
| Body-mounted     | Chair / bed<br>early rehabilitation    | Wearable an-<br>kle exoskeleton<br>with modular fitting | Compact antagonistic<br>SMA actuation              |
| Assistive / gait | Walking assistance<br>daily mobility   | Soft or rigid orthoses<br>for gait support              | Cable, spring, motor<br>or quasi-passive actuation |

**Position of this thesis:** body-mounted ankle rehabilitation for early-stage use in seated or bed-based scenarios, focused on dorsiflexion–plantarflexion with compact SMA actuation rather than gait assistance.